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PLANTER WITH HIGH SPEED SEED DELIVERY APPARATUS

Abstract

An agricultural planting implement includes a number of row units. The row units include one or more seed meters for receiving, singulating, and dispensing seed to the ground such that preferred spacing of subsequent seed is attained. A seed meter provides seeds one at a time to a seed carrier, such as a brush wheel. The brush wheel may move the seeds one at a time to a seed conveyor by directly moving seeds along a curved portion of a seed disc in the seed meter. The seed conveyor may be a flighted belt, and the velocity of the seeds when transferred from the seed carrier may match the velocity of the flighted belt. The seed conveyor conveys the seeds to a position near the bottom of a furrow, and ejects the seeds with little or no horizontal velocity relative to the bottom of the furrow.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation Application of U.S. Ser. No. 18/347,058, filed Jul. 5, 2023, which is a Continuation Application of U.S. Ser. No. 16/949,302, filed Oct. 23, 2020, now U.S. Pat. No. 11,730,079, issued Aug. 22, 2023, which is a Continuation Application of U.S. Ser. No. 16/146,536, filed on Sep. 28, 2018, now U.S. Pat. No. 10,842,072, issued Nov. 24, 2020, which claims priority under 35 U.S.C. § 119 to provisional application U.S. Ser. Nos. 62/565,881, filed Sep. 29, 2017, and 62/596,350, filed on Dec. 8, 2017. The priority patent applications are herein incorporated by reference in their entirety, including without limitation, the specification, claims, and abstract, as well as any figures, tables, appendices, or drawings thereof.

TECHNICAL FIELD

[0002] The present invention relates generally to agricultural implements. More particularly, but not exclusively, the invention relates to an agricultural planter with a seed delivery apparatus for delivering seed from a metering system of a row unit to a furrow created in the ground.

BACKGROUND

[0003] An agricultural row crop planter is a machine built for distributing seed into the ground. The row crop planter generally includes a horizontal toolbar fixed to a hitch assembly for towing behind a tractor. Row units are mounted to the toolbar. In different configurations, seed may be stored at individual hoppers on each row unit, or it may be maintained in a central hopper and delivered to the row units on an as needed basis. The row units include ground-working tools for opening and closing a seed furrow, and a seed metering system for distributing seed to the seed furrow.

[0004] In its most basic form, the seed meter includes a housing and a seed disc. The housing is constructed such that it creates a reservoir to hold a seed pool. The seed disc resides within the housing and rotates about a generally horizontal central axis. As the seed disc rotates, it passes through the seed pool where it picks up individual seeds. The seeds are subsequently dispensed from the seed meter and transported to the seed furrow.

[0005] Seed spacing in the seed furrow is roughly controlled by varying the rotational speed of the seed disc. The most common seed delivery system for delivering seed from the seed disc to the furrow may be categorized as a gravity drop system. In the case of the gravity drop system, a seed tube has an inlet end, which is positioned below the seed metering system. The singulated seeds from the seed metering system drop into the seed tube and fall via gravitational force from a discharge end thereof into the seed furrow. Monitoring systems are commonly used to monitor the operation of the planter. Such systems typically employ a seed sensor attached to each seed tube to detect the passage of seed therethrough.

[0006] However, such a gravity system can affect the seed spacing of the planter. For example, as the spacing of the speed is dependent on the rotational velocity of the seed disc and the gravitational constant, interruptions, forces, or other occurrences acting on the seed can greatly affect the spacing. For example, if the seed bumps against a wall of the seed tube on the way to the furrow; this can cause a delay or a non-vertical fall of the seed. If a preceding or following seed does not experience the same interruption, the seeds could be spaced too close or far from one another.

[0007] Furthermore, as the speed of planting increases, this causes additional problems. Drawing a planting implement through the field at faster speeds increases the speed of deposited seeds relative to the ground, causing seeds to roll and bounce upon landing in the trench or furrow and resulting in inconsistent plant spacing. The adverse agronomic effects of poor seed placement and inconsistent plant spacing are well known in the art.

[0008] Therefore, there is a need in the art for an agricultural planting implement that includes a seed delivery apparatus that aids in delivering seed from a singulating seed meter to a furrow or trench in the field, such that the spacing of adjacent seed is more consistent to increase the yield obtained of the end crop.

SUMMARY

[0009] Therefore, it is a principal object, feature, and/or advantage of the disclosed features to overcome the deficiencies in the art.

[0010] It is another object, feature, and/or advantage of the disclosed features to provide an agricultural planter with a seed delivery apparatus to provide consistent spacing between adjacent seed.

[0011] It is yet another object, feature, and/or advantage of the disclosed features to provide a seed delivery apparatus, mechanism, and/or assembly that will deliver a seed from a seed metering device to the field.

[0012] It is still another object, feature, and/or advantage of the disclosed features to provide a seed delivery apparatus that will provide optimized spacing in a seed furrow.

[0013] It is a further object, feature, and/or advantage of the disclosed features to provide a seed delivery apparatus that will allow for planting with increased speed.

[0014] It is still a further object, feature, and/or advantage of the disclosed features to provide a seed delivery apparatus that provides for seed spacing that will not be influenced by abrupt forces during travel.

[0015] It is yet another object, feature, and/or advantage of the disclosed features to provide a controlled delivery of seed from a seed meter to the ground wherein a seed experiences near zero horizontal velocity relative to the ground, regardless of the velocity of the planter.

[0016] These and/or other objects, features, and advantages of the disclosure will be apparent to those skilled in the art. The present invention is not to be limited to or by these objects, features and advantages. No single embodiment need provide each and every object, feature, or advantage.

[0017] The disclosure relates to various seed delivery systems for providing a desired, equidistant spacing of seed in a field, regardless of the speed of travel of an agricultural planter. Some aspects of the systems can include that the delivery of the seed from a seed meter to a trench or furrow in the ground will not be influenced by factors such as external forces, including the free fall of gravity. Furthermore, at least some of the systems provide setups that provide that the seed will be release with substantially zero relative velocity such that the seed will land softly within a trench or furrow, and will have little to no bounce therein, which will aid in the correct spacing of the seed.

[0018] The disclosure, among other things, relates to a row unit for use with an agricultural implement that includes at least one seed meter having a seed disc. The seed meter has an opening and is adapted to provide one seed at a time to the seed meter opening. A rotating brush in communication with the seed meter opening receives seeds from the seed meter.

[0019] A conveyor in communication with the rotating brush receives seeds from the rotating brush

and the seeds proximate to the ground with a horizontal velocity component near zero relative to the ground. The row unit may include a second seed meter that has a second seed disc. The seed disc may rotate about a seed disc axis that is generally aligned with a direction of travel for the row unit. The rotating brush may rotate about a brush axis that is generally horizontal and transverse to the seed disc axis. The brush axis may be generally perpendicular to the seed disc axis. The row unit may include a comb in engagement with the rotating brush proximate to the conveyor to guide the seeds out of the rotating brush onto the conveyor. The conveyor may have a flighted belt within a conveyor cover. Movement of the flighted belt may be synchronized with a rotation speed of the rotating brush whereby seed exits the rotating brush with a velocity that closely matches the movement of the flighted belt. The seed meters may include a vacuum channel aligned beneath slots on the seed disc for retaining seed on the seed disc, and the vacuum channel may follow a path that moves the seeds radially outwardly on the slots towards the seed meter opening as the seed disc rotates.

[0020] The disclosure also relates to an agricultural planting implement that includes a plurality of row units. Each row unit of the plurality includes a seed meter with a seed exit, a seed carrier in communication with the seed exit for receiving seed from the seed meter one seed at a time; and a seed conveyor in communication with the seed carrier to receive seeds from the seed carrier one at a time at a transfer location remote from the seed meter. The seed conveyor is adapted to move the seed from the transfer location to an ejection location proximate to the bottom of a furrow and is adapted to eject the seed with little or no horizontal velocity relative to the bottom of the furrow. The seed carrier may be a brush wheel. The seed carrier is adapted to release the seeds one at a time at the transfer location with a transfer velocity that closely matches a velocity of the seed conveyor. The seed conveyor may include a flighted belt.

[0021] According to another feature the disclosure relates to a row unit for use with an agricultural implement that has a seed meter. The seed meter has a seed disc that rotates about a seed disc axis. The seed meter has an opening. The seed disc has a front face and a rear face. The front face includes a flat inner portion and a curved outer portion that extends frontwardly from an outer edge of the flat inner portion. An array of apertures is provided in the flat inner portion of the seed disc in a circular pattern spaced radially inwardly from the curved outer portion. Each of the apertures in the array is adapted to retain a single seed. A rotating wheel rotates about a wheel axis. The rotating wheel is located such that an outer portion of the wheel matches and is in close engagement with the curved outer portion of the front face of the seed disc such that as the seed disc rotates it brings a seed retained on one of the apertures into engagement with the outer portion of the wheel whereby the rotating wheel moves the seed outwardly along the curved outer portion to the opening in the seed meter. A conveyor in communication with the opening in the seed meter to receive seeds from the rotating wheel, the conveyor ejecting the seeds proximate to the ground with a horizontal velocity component near zero relative to the ground. The rotating wheel may include a brush. The seed disc axis may be generally aligned with a direction of travel for the row unit. The wheel axis may be generally transverse to the seed disc axis. The brush axis may be generally perpendicular to the seed disc axis. The conveyor may include a flighted belt within a conveyor cover. The flighted belt is synchronized with rotation speed of the rotating brush whereby seed exits the rotating brush with a velocity that closely matches the movement of the flighted belt. The seed meter may include a rear cover facing that covers the rear face of the seed disc and a vacuum source for supplying a vacuum between the rear face and the rear cover to help retain the seeds in contact with the apertures as the seed disc rotates. The row unit may have a gasket that forms an airtight seal between the rear cover and an outer portion of the seed disc, whereby the seed disc rotates relative to the gasket. The gasket may include a leg that extends along the outer wall, such that relative movement of the seed disc and the rear cover towards and away from each other will not break the airtight seal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. 1 is a perspective view of a planting implement according to one embodiment described in the disclosure.

[0023] FIG. 2 is a side elevation view of a row unit for use with a planting implement.

[0024] FIG. 3 is a side elevation view of a row unit including a dual seed meter and seed delivery system according to the present disclosure.

[0025] FIG. 4 shows the seed delivery apparatus of FIG. 3 isolated from the other components of the row unit.

[0026] FIG. 5 is an internal view of a seed meter showing components and a seed path for a seed to move through the meter.

[0027] FIG. 6 shows an inner cover of the seed meter of FIG. 5, including a vacuum channel.

[0028] FIG. 7 is a partial detail view of a seed meter opening and a rotating brush chamber.

[0029] FIG. 8 is a partial detail view of a rotating brush wheel and flighted belt conveyor.

[0030] FIG. 9 is a detail view of a comb used to guide seeds from a rotating brush to a conveyor.

[0031] FIG. 10 is a partial detail view of the transmission elements of a rotating brush and flighted conveyor.

[0032] FIG. 11 is a side elevation view of another row unit showing a unitary seed meter and seed delivery system according to the present disclosure.

[0033] FIG. 12 is a partial detail view of the seed meter, wheel brush, and flighted conveyor of the row unit of FIG. 11.

[0034] FIG. 13 is a side elevation view of a row unit with a seed meter and seed delivery system according to another embodiment of the present disclosure.

[0035] FIG. 14 is a front elevation view of the row unit with seed meter of FIG. 13.

[0036] FIG. 15 is a cross-section side elevation view of the row unit with seed meter of FIG. 14.

[0037] FIG. 16 is a detail close-up of a portion of the cross-section side elevation view of FIG. 15.

[0038] FIG. 17 is a cross-section schematic illustrating features of the seed meter and seed delivery system of FIGS. 13-16.

[0039] FIG. 18 is a partial cross-section isometric view of the seed meter and seed delivery system of FIGS. 13-16.

[0040] FIG. 19 is a detail close-up of the partial cross-section isometric view of the seed meter and seed delivery system of FIG. 18.

[0041] FIG. 20 is a detail partial cross-section view of a portion of the seed meter of FIG. 19.

[0042] FIG. 21 is a partial detail cross section showing the connection and seal between the seed meter wheel and cover at a lower portion of the seed meter.

[0043] FIG. 22 is a partial detail cross-section isometric view showing the connection and seal between the seed meter wheel and cover at an upper portion of the seed meter.

[0044] FIG. 23 is a perspective view of a row unit according to additional aspects of the invention.

[0045] FIG. 24 is a top plan view of the row unit of FIG. 23.

[0046] FIG. 25 is a side elevation view of the row unit of FIG. 23.

[0047] FIG. 26 is a side sectional view of the row unit of FIG. 23 according to line 26-26 of FIG. 24.

[0048] FIG. 27 rear sectional view of the row unit of FIG. 23 according to line 27-27 of FIG. 25.

[0049] FIG. 28 is a perspective view of a seed meter and seed delivery system according to aspects of the invention.

[0050] FIG. 29 is side elevation view of the seed meter and seed delivery system of FIG. 28.

[0051] FIG. 30 is a sectional view of the seed meter and seed delivery system of FIG. 28.

[0052] FIG. 31 is an opposite sectional view of FIG. 30.

[0053] FIG. **32** is a front view of the seed meter and seed delivery system of FIG. **28**.

[0054] FIG. **33** is a rear view of the seed meter and seed delivery system of FIG. **28**.

[0055] FIG. **34** is a top plan view of the seed meter and seed delivery system of FIG. **28**.

[0056] FIG. **35** is a sectional view taken along line **35-35** of FIG. **29**.

[0057] FIG. **36** is a sectional view taken along line **36-36** of FIG. **29**.

[0058] FIG. **37** is a perspective view of a seed disc and singulator according to aspects of the invention.

[0059] FIG. **38** is an elevation view of the seed disc and singulator of FIG. **37**.

[0060] FIG. **39** is a perspective view of a singulator.

[0061] FIG. **40** is a rear view of a seed disc.

[0062] FIG. **41** is a perspective view of a seal for use with a seed meter.

[0063] FIG. **42** is a sectional view of the seal of FIG. **41**.

[0064] Various embodiments of a seed delivery system and related components are described in detail with reference to the drawings, wherein like reference numerals represent like parts throughout the several views. Reference to various embodiments does not limit the scope of the invention. Figures represented herein are not limitations to the various embodiments according to the invention and are presented for exemplary illustration of the invention.

DETAILED DESCRIPTION

[0065] FIG. **1** shows an agricultural implement **10**, in this case, an agricultural planter. The planter **10** is usually attached to and pulled by a tractor. However, it should be appreciated that other equipment and/or vehicles may move the implement **10**. For purposes of the present disclosure, the implement **10** will be referred to as a planter.

[0066] The planter **10** includes a tongue **14** having a first end **16** and an opposite second end (not shown). The tongue **14** includes a hitch **18** at the first end **16**, with the hitch **18** being connected to the tractor. At the opposite end of the tongue **14** is a central tool bar **22**. The tongue **14** may be a telescoping tongue with components capable of being inserted into one another such that the implement **10** is a front folding style implement. However, the present invention is not to be limited to such front folding style implements and is to include any such implement for use in the agricultural industry.

[0067] As shown in FIG. **1**, central hoppers **24** are positioned at the central toolbar **22**. The hoppers **24** are configured to store seed, fertilizer, insecticide, or other types of material for use in farming. The hoppers **24** may both contain the same material, or could contain separate materials. The use of the central hoppers **24** allows for a large amount of material to be added and stored at a centralized location. However, the invention also contemplates the use of one or more hoppers positioned at each of the row units **34** for providing seed to be planted at the row units, as is shown in FIG. **3**. When central hoppers **24** are used at the central toolbar **22**, it should be appreciated that the central hoppers will be in fluid communication with each of the row units **34**. This can be done by use of separate hoses to each of the row units, or fewer hoses that include splitters, wherein the hose is split to provide seed or other material to more than one row unit. Also connected to the central toolbar is a plurality of central wheels, which may be known as transport wheels **26** extending generally downwardly from the central toolbar **22**. The wheels **26** contact the ground and support the central hoppers **24**. The wheels stabilize the implement **10** and are the wheels that contact the ground when in a working position or a transport position, e.g., if the implement **10** is a front folding implement such that the wings **28**, **30** are folded forward with wing wheels **32** not contacting the ground.

[0068] Extending generally from both sides of the toolbar **22** are first and second wings **28**, **30**. The wings **28**, **30** are generally identical and mirror images of one another. Therefore, only one wing will be described with the understanding that the other wing will be generally the same configuration. The first wing **28** includes a bar **29**. Mounted to the bar **29** are a plurality of row units **34**, as well as a plurality of wheels **32**. The wheels **32** are configured to contact the ground.

The row units **34** may be seeders, fertilizers, insecticide sprayers, or other dispensers, discs, or plows. The wings **28, 30** may also include at least one fold cylinder and a down force cylinder. It is further contemplated that multiple down force cylinders be used with an implement having more sections. The fold cylinder(s) is configured to fold the wings to a position wherein the first and second wings **28, 30** are generally adjacent the tongue **14** of the implement **10**.

[0069] FIG. **2** is a side elevation view of a row unit **34**, and more specifically, a seeder including a singulating seed meter **36**. The row unit **34** includes a seed meter **36**, furrow opener **38**, row hoppers **39**, and furrow closer **40**. The row unit **34** also includes standard features such as frame **41**, linkage **42**, mount **43** for mounting to the planter **10**, gauge wheels **44**, and depth control mechanism **45**. The gauge wheels **44** and depth control mechanism **45** work together to control the depth of the furrow or trench created by the opener **38**. Furthermore, the row hopper **39** is connected to a seed supply, such as the central hoppers **24**, which can provide the seed meter **36** of each row unit **34** with seed to be planted. Alternatively, the row hopper **39** may be a self-contained unit unconnected to a central hopper.

[0070] FIG. **3** is a cut-away side elevation view of another embodiment of a row unit **34** according to exemplary aspects of the disclosure. The embodiment of FIG. **3** is a dual meter seed delivery apparatus **50** that includes two seed meters **36** that can contain different hybrids of the same crop or even seeds for different crops. Typically, only one of the seed meters **36** would be operating at any one time, but there could be overlap and/or operation of both meters contemporaneously, simultaneously, offset from one another, or otherwise in coordination with one another. The meters **36** would turn off and on depending on which hybrid or crop is needed to be planted in each portion of the field. Both meters **36** could be run simultaneously for a high population or to plant two different crops at the same time. In operation, seed would be removed from one (or both) disc of a meter and would be in contact with the rotating brush **54**, such as at the left portion of the brush **54** as shown in FIG. **3**. The brush **54** would be rotating in a counterclockwise direction when looking at FIG. **3**. The brush **54** would then pass the seed to the flighted belt **58**, which would take the seed to the ground. As would be understood, the seed would move in an “S” shape from the brush to the belt and then to the ground, as it would be delivered towards the left of FIG. **3** when released from the belt **58**. Other exemplary embodiments of the invention utilize a single meter system, for example as shown in FIGS. **10-11**.

[0071] FIG. **4** shows the seed delivery apparatus **50** isolated from the row unit **34** shown in FIG. **3**. As seen in FIG. **4**, the seed delivery apparatus **50** includes a vacuum connector **52** for applying a vacuum to the seed meters **36**. It is also noted that the vacuum could be replaced with positive pressure, making the seed meter an air seed meter regardless of the pressure type. Still further, as will be understood with respect to additional figures, any of the embodiments of row units, seed meters, seed delivery systems, and/or any combination of the same can include dedicated or otherwise integrated pressure systems. Such pressure systems are shown and described in co-owned U.S. Pat. No. 9,763,380, issued on Sep. 19, 2017, the contents of which are hereby incorporated by reference in their entirety and for all purposes.

[0072] The operation of the seed meters **36** is described in more detail below with reference to FIG. **5**. The essential feature of the seed meters **36** is that they present seeds, one at a time, to a rotating brush **54** that is in communication with the seed meters **36**. The rotating brush **54** moves the seeds to a conveyor **56**. The conveyor **56** delivers the seeds proximate to a furrow where the seeds are ejected from the conveyor **56** with a horizontal velocity component that substantially and reciprocally matches the ground speed of the row unit **34**, such that the horizontal velocity of each seed relative to the ground is zero, or nearly zero. In other words, the seeds fall more or less straight down without forward momentum being imparted by the direction of travel of the planter. In the embodiment of FIG. **4**, the conveyor **56** includes a flighted belt **58** within a case **60**. The belt **58** is made of a resilient material that includes equally spaced flights **62** extending normally from the surface of the belt **58**. In operation, the spaces between adjacent flights **62** act as seed

receptacles **64** such that a single seed will be located within any one receptacle to be transported from the brush **54** to a release point **66** near the bottom of the furrow, typically at or below ground level. For example, the release point may be about 1.5 inches from the bottom of the furrow. [0073] FIG. 5 shows a detail cross-section view of a seed meter **36** according to an exemplary embodiment of a dual-meter system. Each seed meter **36** includes a rotating disc **68** within an outer case **70**. A front surface **69** of the disc **68** is visible in FIG. 5. Seed retaining slots **74** are arrayed along the outer portion of the disc **68**. The slots **74** extend angularly relative to a corresponding radius of the disc **68**, such that the inner portion of each slot **74** leads the outer portion of the slot as the disc rotates (in a clockwise direction as seen in FIG. 5). Paddles **76** extend from the perimeter of the disc **68**. The paddles **76** are oriented generally transversely perpendicular to the plane of the disc **68**. The outer edges of the paddles **76** are close to the inner surface of the outer case **70** such that no seeds can fit between the paddles **76** and the inner surface of the outer case **70**. Outer and inner singulator blades **78** & **80** are provided on the inner compartment wall **73** and outer case **70** respectively. An inner cover **82** is provided between the disc **68** and a vacuum chamber that is located between the meters **34**. As seen in FIG. 6, the inner cover **82** includes a vacuum channel **84** that forms a vacuum path aligned under the slots **74**. The vacuum channel **84** tracks radially outwardly relative to the disc **68** between about 4 o'clock and 6 o'clock as viewed in FIG. 5. In operation this outward tracking of the vacuum channel **84** helps move the singulated seeds retained on the slots **74** outward towards an opening **86** at the bottom of the meter **34**.

[0074] As shown in FIG. 5, a seed reservoir **88** is formed by internal wall **72** and inner compartment wall **73**. As further shown in FIG. 5, a seed pool would be formed generally in the lower left quadrant of the disc **68**. As the disc **68** rotates in a clockwise direction individual seeds from the seed pool adhere to the slots **74**. The singulator blades **78** & **80** bump off any excess seeds so that by the time a slot **74** rotates past the singulator blades **78** & **80** the slot **74** will contain only one seed in alignment with the vacuum channel **84**. As the disc **68** continues to rotate, the vacuum channel **84** causes the seed to move radially outwardly on the slot **74** starting at about the four o'clock position causing the seed to pass through the opening in the inner compartment wall **73**. The trailing paddle **76** pushes the seed through a seed loading opening **86** in the outer case **70** where the seed is picked up by the rotating brush **54**. The seed loading area is shown in detail in FIG. 7. Fins **88** span between adjacent paddles **76**. The fins support the paddles **76**, but more importantly match the contour of the seed loading area to permit the brush wheel to pass by without interfering with any seeds that might have been placed in the brush upstream, and also to retain the seeds within the brush cavity.

[0075] The rotating brush **54** rotates about an axis that is generally (substantially) normal or perpendicular to the axis about which the disc **68** rotates. In other embodiments, the axes could be transverse at angles other than 90 degrees. In some embodiments it is preferred that the axis about which the brush **54** rotates should be generally perpendicular to the direction of travel of the row unit **34** so that the brush **54** imparts a motion to the seeds that is parallel to the direction of travel of the row unit **34**. The rotating brush **54** has resilient bristles that gently retain the seed, and the seed is carried within the brush as it rotates. According to at least one embodiment, the rotating brush **54** includes nylon bristles and is formed by mounting a strip of nylon bristles on a circular hub. The outer diameter of the brush according to one embodiment is about four inches; however other dimensions are feasible depending upon the need. As depicted, the bristles have a wave or saw tooth pattern; however, this not a necessary feature of the brush.

[0076] As shown in FIG. 7, the paddles **76** have a small profile as projected relative to the direction of travel of the brush **54**. Furthermore, the speed of the bristles relative to the paddles **76** in the direction of travel of the bristles is much greater than the speed of the paddles **76** relative to the bristles in the direction of travel of the paddles **76**. Accordingly, the paddles **76** will pass through the bristles of the rotating brush **54** without significant lateral displacement or deformation of the bristles. According to one embodiment the bristles may be moving about ten times as fast as the

paddles **76** at the interface between the paddles **76** and the brush **54**.

[0077] The rotating brush **54** acts as a seed carrier to carry seeds from the seed meters **36** to the conveyor **56**. According to an aspect of some embodiments, the rotating brush **54** and the flighted belt **58** are substantially synchronized so that the velocity of a seed as it leaves the brush **54** closely matches the velocity of the flighted belt **58** so that the seed transitions smoothly to a seed receptacle **64** between adjacent flights **62** with minimal jarring. As shown in FIG. **8**, the transition of the seed from the brush to the conveyor belt **58** occurs at or near the top of the conveyor **56**, where the belt **58** wraps around a tensioning roller (drive pulley) **90**. The wrapping of the belt **58** angles the flights **62** away from each other creating a larger opening to the seed receptacle **64** between the flights **62** giving a larger window to transition the seeds accurately one per receptacle **64**.

[0078] As shown in FIG. **9**, a comb **92**, or similar structure, may be mounted to the conveyor case **60** at the transition area between the brush **54** and the conveyor **56** to facilitate removal of the seeds from the brush **54**. Specifically, the comb **92** may include individual tines **94** that extend into the brush **54**. The tines are shaped to deflect and guide the seed downwardly out of the brush **54** onto the moving flighted belt **58**.

[0079] As shown in FIG. **10**, the synchronization of the brush **54** with the flighted belt **58** may be accomplished by interconnecting the drive mechanisms for the brush **54** and the flighted belt **58**. For example, the brush **54** may have gear **96** that is driven by gear **100**, and tensioning roller **90** may be driven by gear **98**. Because gears **98** and **100** are meshed, they cause the rotation of the tensioning roller **90** and the brush **54** to be synchronized. One of gears **98** or **100** is driven and provides impulse that corresponding drives the other of the gears **98** or **100**. The speed of the gears **98** and **100** is controlled mechanically, or electronically, based upon the ground speed of the planter and the desired spacing of the seeds. Accordingly, as the ground speed increases, the speed at which the brush **54** rotates and the belt **58** moves increases to eject seeds at a higher speed to maintain a uniform spacing and assure that the seeds are falling straight down with little or no forward or rearward momentum relative to the ground. Each of the components, the seed disc **68**, rotating brush **54** and the conveyor **56** may be driven by individual electric motors that are coordinated by a central control or processing unit (not shown) that receives input regarding the ground speed of the unit and desired spacing of the seeds. Alternatively, components may share driving force from one or more inputs such as electric motors, internal combustion motors, or motion driven linkages.

[0080] Similarly, the speed of the rotation of the disc **68** in the seed meters **36** is also proportional to the ground speed of the planter **10** (and desired planting population) so that seeds are being provided to the brush **54** at the proper rate. The disc **68** may be mechanically connected to wheels on the ground to assure that the speed of the disc **68** is proportional to the ground, or electronic sensors may be used to set the speed of the disc **68**. Still further, GPS, tractor speed calculations, or the like, may indicate and/or otherwise provide the ground speed for the rotational speeds to use to attempt to substantially match such that the seed is released with zero relative velocity. The ground speed could be the tractor, the planter, portions of the planter (e.g., at the row units), or some combination thereof. The disc **68** may be driven by electronic step motors or other known devices for driving rotation.

[0081] FIGS. **11** and **12** show a row unit **234** that is similar to unit **34** described above, but utilizes a single seed meter **236**. A seed delivery apparatus **250** includes a seed meter **236** that presents seeds, one at a time, to a rotating brush **254** that is in communication with the seed meter **236**. The rotating brush **254** moves the seeds to a conveyor **256**. The conveyor **256** delivers the seeds proximate to the bottom of a furrow where the seeds are ejected from the conveyor **256** with a horizontal velocity component that reciprocally matches the ground speed of the row unit **234**, such that the horizontal velocity of each seed relative to the ground is zero, or nearly zero. Similar to the dual-meter embodiment of FIG. **4**, the conveyor **256** includes a flighted belt **258** within a case **260**.

The belt **258** is made of a resilient material that includes equally spaced flights **262** extending normally from the surface of the belt **258**.

[0082] In the single meter embodiment of FIGS. **11** and **12**, the seed meter **236** includes a rotating disc **268** that rotates about an axis that is aligned with a direction of travel for the row unit **234**, but inclined relative to horizontal. The seeds are provided to the rotating brush wheel **254** at a lower portion of the seed meter **236**. However, unlike the dual-meter design described above, rather than using paddles **76** to move the seed into the brush **254**, the single seed meter **236** relies upon the path of the vacuum channel, as well as gravity and the momentum of the seed to move the seed into brush **254**. Once the seed is received in the bristles of the brush **254**, the brush **254** carries it a short distance to the conveyor **256** where the seeds are transferred to the conveyor belt **258**. A comb **292** helps guide the seeds on to the belt **258** between the raised flights **262**. The seed is preferably moving with the same velocity as the flights **262** when it is released from the brush **254** to assure a smooth transition to the belt **258**. The seed then travels down the length of the conveyor with one seed between each adjacent flight **262** under the force of gravity. This does not necessarily mean a seed will be between every two flights, but instead, a seed will be temporarily housed between two flights, the inner portion, and an end wall of the belt housing. The equal spacing of the flights **262** assures that the seeds will be equally spaced as they are ejected reward from the bottom of the conveyor **256** at a velocity that offsets the forward speed of the planter.

[0083] FIGS. **13-22** illustrate another embodiment of this disclosure. A row unit **300** is shown in FIGS. **13-16**. The row unit **300** includes many similar features as described herein. A mounting bracket **302** permits attachment to an implement bar. A linkage **304** connects the mounting bracket **302** with a frame **306** that supports the working elements of the row unit **300**. The frame **306** is connected to a furrow opener **310** in the form of disc blades or the like. A furrow closing device (not shown) is mounted to a furrow closing adjustment mechanism **318** at the trailing end of the unit **300**. Gauge wheels **312** are provided in close proximity to the furrow opener **310** to control the depth of the furrow opener **310**. An adjustment mechanism **314** is provided to adjust the relative position of the gauge wheels **312** to the furrow opener **310**. Also mounted to the frame **306** is a seed delivery apparatus that includes seed meter **308** and a seed conveyor **316**.

[0084] Further details of the seed delivery apparatus of the row unit **300** can be seen in the cross-section views of FIGS. **15** and **16**. The seed meter **308** includes a rotating seed disc **322**. The seed disc **322** includes several apertures **324** that each retain a single seed (not shown) as the seed disc **322** rotates, in a similar manner to the embodiments described above. However, in the seed delivery apparatus of unit **300**, a rotating wheel, such as rotating brush **320** is provided in close engagement with a front face (seed side) of the disc **322** to disengage and/or otherwise remove the seed from the aperture **324** and move the seed towards the conveyor **316**. More particularly, the brush **320** moves the seed radially outwardly (relative to the seed disc) along a curved portion **326** of a front face (seed side, which is opposite the vacuum side of the disc) of the rotating seed disc **322**. As described in the embodiments above, the brush wheel **320** accelerates the seed to closely match the speed and direction of a conveyor belt **328** within the conveyor **316** that takes the seed to a release position very near the bottom of the furrow. In contrast to the embodiments described above, in this embodiment, the wheel brush **320** engages the seed directly from the face of the seed disc and moves it towards the conveyor **316**, as compared to the embodiments described above wherein the wheel brush is located tangentially to the seed disc and receives the seed after has disengaged from seed aperture.

[0085] FIG. **17** shows a schematic that illustrates some of the features of a seed delivery system **301** that incorporates the concepts of the system used in row unit **300**. The seed disc **322** in the seed meter **308** rotates about a seed disc axis **330**. The seed meter **308** includes structure for causing the rotation of the seed disc **322** that is not shown, but is well-known (e.g., electric motor operatively connected to the disc). The seed disc **322** has a front face **332** (aka the seed side) and a rear face **334** (aka as the vacuum or pressure side). The front face **332** of the seed disc **322** is generally flat

surrounding the seed disc axis **330**, though this geometry is not generally critical. An outer portion **326** of the seed disc **322** is curved to correspond with the radius of the rotatable wheel **320** that rotates about a wheel axis **321**. The curvatures need not match in all embodiments. The apertures **324** are provided through the seed disc **322** at or near where the curved portion **326** begins. Individual seeds **336** are retained on front face **332** of the seed disc **322** at the apertures **324**. The seeds **336** are retained in place by a pressure differential across the apertures **324**, which could be a positive or negative (vacuum) pressure. According to the embodiment shown, a partial vacuum is created between the rear face **334** of the seed disc **322** and a rear cover **338**. A singulating mechanism (see FIG. 22) may be used to assure that only a single seed **336** is associated with each aperture **324**. A gasket **340** surrounds an outer wall **341** of the rotating seed disc **322** and provides a generally air tight seal between the seed disc **322** and the rear cover **338**. While the gasket **340** is positioned at the outer wall in the figure, it is noted that it could also be placed at the inner side of the seed disc outer wall, such as shown in other embodiments. Therefore, it should be appreciated that the gaskets could be placed outside of or inside of the outer wall/edge of the seed disc such that it will create a substantially full pressurized zone of the disc. A front cover **342** encloses the seed meter **308** and snap fits with the rear cover to hold the gasket **340** in place. The seed disc **322** can rotate against a leg of the gasket **340**. Because the gasket **340** extends up a portion of the outer wall **341**, the seal between the seed disc **322** and the rear cover is not broken by small relative movement of the rear cover and the seed disc **322** relative to each other.

[0086] In operation, the seed disc **322** rotates continuously. As the apertures pass through a seed pool (not shown), seeds **336** will adhere to the apertures **324** and be singulated by passing through a singulator structure **348** (see FIG. 20). Eventually each seed **336** on an aperture **324** reaches the rotating brush wheel **320**. The rotating brush wheel **320** engages the seed **336** and moves it radially outwardly off of the aperture **324** along the curved surface **326** of the seed disc **322**. The wheel **320** accelerates the seed to match the speed and direction of the outer portion of the wheel **320** and provides the seed **336** at an opening **344** in the seed meter that is adjacent to the conveyor at a velocity that closely matches the speed and direction of the conveyor belt **328**. Because the wheel **320** is moving much faster than the seed disc **322** (on the order of ten times faster at the apertures **324**) there is little circumferential movement of the seed **336** relative to the brush **320** due to momentum and friction with the rotating seed disc **322** over the short distance between the aperture **324** and the conveyor **316**. In that regard it can be beneficial for the front face **332** of the seed disc **322** to be a low friction surface to both reduce friction forces applied to the seed **336** as it is moved by the brush **320** which could cause unwanted lateral movement of the seed and in extreme cases potentially damage to the seed itself.

[0087] FIGS. 18-22 show various views of various features of the seed delivery system **301**. In FIGS. 18 and 19 a comb structure **346** is shown that assures that the seeds do not stick in the brush **320** and are transferred to the conveyor belt **328**. As best shown in FIGS. 19 and 21, the gasket **340** includes a flexible leg that extends along the outer wall **341** of the seed disc **322**. Again, it is noted that the gasket **340** could also be placed at the inner wall or inner side of the wall as well. The front cover **342** and the rear cover **338** include projections that capture the gasket **340** to hold it in place and to effectively stiffen the leg of the gasket **340** by limiting its free flexing portion. The singulator structure **348** is best seen in FIG. 20. The structure is fixed to an inner surface of the front cover. As seeds on the circular array of apertures **324** pass through the singulator structure **348** excess seeds are knocked off, leaving a single seed on each aperture. The singulator can take many forms, such as a single or multi-blade singulator (see, e.g., U.S. Pat. No. 9,277,688, hereby incorporated by reference in its entirety), knock-offs, wheels, or brushes, such as that shown in FIG. 39 of the present disclosure. The type of singulator should not be limiting to the invention.

[0088] FIGS. 23-27 disclose additional views of a row unit **400** according to and including aspects of the invention. Similar to that previous disclosures of row units, the row unit **400** includes a plate **402** or other member for attaching the row unit **400** to a toolbar of a planting implement. Extending

generally from the faceplate **402** is a linkage **404**, which allows for some vertical movement of the row unit **400** relative to other row units and components of the planting implement. While not shown, a system, apparatus, and/or assembly for providing down and/or up force for the row unit may also be included for use with the row unit. Such a system is shown and described in U.S. patent application Ser. No. 16/047,236, which is hereby incorporated by reference in its entirety. A frame **406** is also included. The row unit **400** includes opening elements **408**, which are in the form of opening or coulter wheels in the figures. Gage wheels **410**, a depth adjustment system **412**, and a sensor **418** are included to adjust the depth of the furrow created and to sense and adjust an amount of supplemental down and/or up force for the row unit.

[0089] Connected to the frame **406** is a hopper attachment **414**, which may also be referred to as a mini hopper. The attached **414** includes, in part, a lid portion, a receptacle portion, and a conduit attachment **416**. The conduit attachment **416** can be attached, via delivery system, to one or more bulk hoppers of the planting implement to receive on-demand amounts of seed for the row unit. The seed can be stored, at least temporarily, at or in the attachment **414**, where it can then be fed into a seed meter seed pool for singulation and delivery to the ground.

[0090] FIGS. **26** and **27**, which are sectional views of the row unit **400**, show additional aspects of the invention, which include, in part, the seed meter **420** and the seed to ground delivery system **450**. The seed meter **420** and the seed delivery system **450** are similar in nature to that shown and described with respect to FIGS. **13-22**. The seed meter will receive seed via the conduit **416** and temporarily store the seed at a seed pool **423** within a housing **422** of the meter **420**. A disc **430** rotating within the housing **422** will interact with the seed in the pool **423**, and a pressure differential at seed apertures in and through the disc will cause the seed to adhere at the apertures of the disc **430**. It is noted that the entire second side of the seed disc **430** will be pressurized, and there will be no areas where the pressure difference is cut-off. The seed will travel on the disc **430** until it is dislodged by a combination of a seed knock-off **424** and a brush wheel **452**, which are on opposites side of the seed disc **430**, as will be disclosed herein. The seed will be transported via the brush wheel **452** at a higher rotational velocity than the rotating speed of the disc **430** to a point where the seed will be delivered towards a belt **454**. The belt **454** includes spaced flights **455**, which will provide a controlled delivery towards a release point **462**. The belt **454** will have a velocity that is synced with the ground speed at the row unit **400** such that the seed will be released from the belt **454** and delivery system **450** with a horizontal velocity component that is equal to said ground speed in an opposite direction so that the seed is release with zero relative velocity, which will mitigate bounce, roll, or other movement of the seed when it contacts the ground.

[0091] FIG. **27** is a rear sectional view of the row unit **400** showing the second or pressurized side of the seed disc with the seed knock-off member **424**. As shown, the knock-off member **424** is a rotating member with spikes or other elongations extending outward and spaced to substantially align with corresponding seed apertures of the seed disc. This would make it such that the elongations extend at least partially into the seed apertures to interact with the seed to dislodge the seed thereat to aid in the removal of the seed by the brush wheel **452** that is rotating substantially transversely to the rotating seed disc.

[0092] FIG. **27** also shows the location of the belt **454** relative to the opening wheels **408**. The belt housing **456** will be at least partially between the opening wheels **408** to place a seed dispersed therefrom in the created furrow as close to creation as can be to mitigate the movement of the seed dropped therein before the furrow is closed. FIG. **27** also shows how the seed meter **420** will be positioned substantially perpendicular to the direction of travel of the row unit **400**, while the brush wheel **452** and the belt **454** will be substantially aligned with the direction of travel.

[0093] FIGS. **28-29**, as well as FIGS. **32-33**, show various views of the seed meter **420** and the seed to ground delivery system **450**, including the housings thereof. The seed meter **420** includes a disc **430** within a housing **422**. The housing **422** may be a one-piece or multi-component housing in which the components are attached to one another, such as by snapping together, clasping, of

otherwise affixing temporarily or selectively to one another. For example, in some embodiments, the housing **422** may include a pressure side **427** and a seed side **429**. The pressure side **427** can include a connection for a pressure source, such as a vacuum, fan, blower, or the like. In the embodiments shown, the pressure source is a dedicated or otherwise integrated fan **428** that is positioned on the housing **422**. The integration of the fan **428** at the seed meter provide numerous advantages, such as individual control for each seed meter of each row unit, greater efficiency, reduction of hoses/conduits for the planter, etc. Additional advantages and description of integrating a fan **428** to the meter housing **422** are shown and described in U.S. Pat. No. 9,763,380, which is hereby incorporated by reference in its entirety. The fan **428** can include a rotating member rotating at a high speed to create a pressure differential at the seed meter, and can convey the pressure to within the housing via a cutout, aperture, conduit, or the like. The fan **428** can be electrically connected to a source that provides electric power thereto in order to operate. Additionally, the fan can be connected to a processing unit, central processor, or other computing member to provide commands and/or control of the fan at each of the seed meters at each of the row units.

[0094] Also shown at the pressure side of the housing **422** is an electric motor **426**. The motor **426** is used to provide rotational power to the seed disc **430** inside the housing. As will be understood, the disc **430** includes gear teeth **435** that will interact with an output shaft of the motor **426**, wherein rotation of the output shaft will result in corresponding rotating of the disc. The electric motor **426** will also be connected to the central processor/control to provide a rotational velocity that is based, in part, on the seed type, population input, ground speed at the row unit, spacing, and other inputs related to planting. The inputs could be inputted to, reviewed, and updated via a central processor and display, such as is disclosed in co-owned PCT Application No. PCT/US2017/064246, which is hereby incorporated by reference in its entirety.

[0095] Opposite the pressure side **427** of the housing **422** is the seed side **429**. The seed side includes a pool **423** or a passage for seed to be delivered to a pool for the seed meter. The housing members may comprise a rigid polymer such as plastic or the like, and can be configured to reduce static energy of the components, such as by including one or more grounds.

[0096] Positioned adjacent to and operatively connected to the seed meter housing **422** is the seed to ground system **450** and components. The components include the brush wheel **452** and housing **453** thereof. As will be understood, the brush wheel comprises a plurality of bristles extending from an axis that are pliable and resilient to be deformed to receive a seed to transfer the seed from the seed disc **430** to the belt **454**. The housing **453** can also be a rigid material, such as a plastic.

[0097] Attached to and extending from the brush wheel housing **453** is the housing **456** for the belt **454**. The belt housing **456** is an elongated member, comprising a rigid material, that is used to house the belt **454** and other components therein. The belt housing **456** can be unitary or multi-component, in which the components are attached to another to allow selective access to the interior thereof. An opening **462** is positioned at or near the bottom of the housing **456** to coincide with the release point wherein the seed is released from the belt and directed towards the furrow for planting.

[0098] While not explicitly shown, one or more sensors will be associated with the housing **456** of the belt **454** to sense the occurrence of seed passing therethrough, the speed of the belt, and other aspects of the movement of the belt and/or seed within the housing **456**. The sensor(s) can be placed generally anywhere along or on the belt housing **456** in which the sensor is able to sense the belt and/or seed moving therein.

[0099] Additional components shown in the figures include a motor **463** and gearbox **464**. The motor **463** is an electrical motor, such as an DC brushless motor. The motor **463** is used to provide power to rotate the brush wheel **452** and the belt **454**. The power is transferred from an output shaft of the motor to the brush wheel **452** and belt **454** via the gearbox **464**. The gearbox **464** includes gears to receive and transfer the rotational output of the motor **463**. The gears may be numbered to

receive direct input from the motor or to receive translated movement from a corresponding gear, and can be sized and spaced to generate a desired output (i.e., rotational velocity), as is known. The motor **463** can be connected to the central processor or controller to provide for a speed to the system **450** based upon the ground speed, population, and other inputs, and can be set such that the seed being released at the release point **462** can have the horizontal component that is substantially equal to and opposite of the ground speed such that the seed experiences zero net velocity. The gears and motor will provide such a speed output.

[0100] FIGS. **30** and **31** are opposite sectional views of the seed meter **420** and the seed delivery system **450** as shown and described. As shown in the figures, the seed meter **420** includes additional components, such as a bearing member **470** to aid in the rotation of the disc **430** within the housing. The disc **430** is also shown to have a first side **431**, which may also be referred to as the seed side, and a second side **432**, which also may be referred to as the vacuum or pressurized side. The seed side **431** is the side that engages and interacts with seed, and the pressurized side **432** is the side that is being pressurized to create the pressure differential at the seed apertures **434** in and through the disc **430**. As will be understood, the disc **430** includes a generally planar inner portion **438** and a curved outer portion **439**. The apertures **434** are positioned substantially at the intersection of the inner and outer portions, where the disc begins to curve. The curvature of the outer portion **439** of the disc **430** coincides generally with the brush wheel **452** such that the brush wheel is able to rotate along with the curvature of the profile of the disc. The point where the brush wheel **452** and the disc **430** interact is shown by numeral **474** in the figures.

[0101] As disclosed, the brush wheel **452** includes a plurality of outwardly extending bristles that are used to engage a seed from the disc and to speed up and transfer the seed to the belt **454**. This occurs generally at the location **475** as shown in the figures.

[0102] Additional elements of the belt **454** include, but are not limited to, a lower roller **458** and an upper roller **459**. The belt **454** comprises an inner surface **457** with flights **455** spaced and extending from the inner surface **457**. The inner surface **457** extends around the upper and lower rollers **458**, **459**, and can be tensioned by a tensioner **460**. The upper and/or lower roller can be connected to the gearbox **464** and motor **463** to provide rotational velocity to the belt **454**, which can correspond with the ground speed of the row unit **400** to provide population selected spacing and planting.

[0103] Still further, the sensor **472** can be associated with the belt **454**, such as at the belt housing **456**, to sense the movement of the belt and to sense seed characteristics as the belt moves. For example, the sensor can be used to determine the presence of a seed, the proximity of one seed to a subsequent seed, multiples, skips, or any other information that corresponds to planting. The sensor can provide feedback to the processor, controls, and/or user to determine the efficiency and accuracy of the planting system to determine if errors are occurring so that they can be troubleshooted and corrected to plant the seeds with the spacing and population desired. Additional sensors may also be positioned along the belt, including on the opposite side to ensure that the seed has been released from the belt and is not continuing to travel with the belt.

[0104] Operation and travel of the seed can be shown best with regard to FIG. **30**. Seed is provided to the seed meter **420** and pooled at the seed pool **423**. A disc **430** rotating in the meter **420** is pressurized to create a pressure differential at the seed apertures **434** in and through the disc. The seed is therefore adhered to an aperture **434** as the aperture passes through the pool **423**. The disc **430** rotates through a singulator **500** to ensure that a single seed is positioned at the aperture **434**. The disc **430** continues rotation towards the brush wheel **452**, which is located at an approximately 6 o'clock position **474** of the meter **420**. At or near this position, on the second side **432** of the disc **430** is a knock-off member **424** that rotates and includes elongations extending at least partially into the apertures **434**. The knock-off member **424** aids in dislodging the seed from the aperture **434** at or near the location of the brush wheel **452** to aid in the removal and transfer of the seed from the disc **430** to the brush wheel **452**. The brush wheel **452** at least partially captures, grabs, or

otherwise moves the seed from the disc **430** and towards the belt **454**. The brush wheel **452** is moving at a much higher rotational speed (e.g., up to 10 times faster) than the disc **430** to begin corresponding to the ground speed of the row unit and/or planter. The bristles of the brush wheel **452** aid in moving the seed.

[0105] The brush wheel **452** moves the seed towards the belt **454**, and transfers the seed to the belt at an approximate 5 o'clock position as shown in FIG. **30**. Thus, the brush wheel does not move the seed very far. The seed is then transferred to a location between adjacent flights **455** of the belt **454** at the transfer point **475**, where the belt **454** begins moving the seed. The belt **454** is being moved at a speed to correspond with the ground speed, and is controlled by the motor **463**, gearbox **464**, and rollers **458**, **459**. The seed will pass at least one sensor **472** to acquire information about the movement before continuing along the elongated path indicated by the arrow **478** in FIG. **30**. The movement will continue towards the release point **462** at or near the bottom of the elongated belt **454**. The seed is released from the belt **454** with a horizontal velocity component that is substantially equal and opposite in direction to the travel of the row unit **400** to drop the seed with zero relative velocity, which will mitigate bounce, roll, or other movement of the seed in the furrow. Thus, the seed has been planted using the seed meter **420** and seed delivery system **450** of the embodiments.

[0106] FIGS. **34-36** are additional views of the seed meter **420**. FIG. **35** is a partial sectional view that shows portions of the first side **431** of the seed disc **430**. As will be understood, the disc **430** includes a plurality of radially spaced apertures **434** creating a seed path. The apertures extend in and through the disc **430**. Adjacent to and in angular relation to the apertures are corresponding seed channels or pockets **436**. The channels **436** engage the seed in the seed pool **423** (location shown in FIG. **35** to be at an approximate 8 o'clock position) to agitate and urge the seeds towards the apertures. The channels are indents in the face of the first side **431** of the disc **430**, but do not extend through the disc **430**. Also shown in FIG. **35** is a singulator **500**, which will be disclosed in more detail herein. The singulator **500** is used to singulate seeds, i.e., ensure that a single seed is positioned at each aperture **434**.

[0107] FIG. **36** is an opposite view of FIG. **35**, with the housing member **427** removed to show the interior of the housing **422**. The figure shows the second or pressurized side **432** of the seed disc **430**, which includes the apertures **434** that extend through the disc. Also shown in FIG. **36** is the gear teeth **435** that engage and interact with the output shaft of the motor **426** to provide rotation to the disc **430**. The gear teeth **435** are shown to be positioned radially from the axis of the disc **430** between the axis and the outer edge of the disc. The teeth **435** are outwardly facing (opposite the axis). Thus, the shaft will not interfere with or get in the way of the seed path or other components of the meter **420**. The knock-off member (also referred to as an ejector) **424** is also shown in the figure. The ejector **424** is a rolling type ejector wheel that aids in seed removal from the disc. As disclosed, the knock-off member **424** includes a rotatable member with outwardly extending portions that correspond with the seed apertures **434**. The rotating member is connected to the housing **422** via a hinge **425** to allow for axial movement of the knock-off member **424** relative to the disc **430**. The member **424** can be biased towards the disc as well.

[0108] FIGS. **37** and **38** are views of the seed disc **430**, and in particular, the first or seed side **431** of the seed disc **430**. As disclosed, the seed disc **430** includes an axis and a plurality of seed apertures/cells **434** radially spaced from the axis and forming a seed path. The apertures **434** extend in and through the seed disc **430**. Adjacent to the apertures **434** are corresponding seed channels **436**, that are angled and extend partially into the disc. The channels **436** are included to agitate and to urge the seed from the seed pool towards the seed apertures **434**. The channels **436** may have an angled front wall and internal end wall, and a rear wall that is substantially normal to the face of the disc. The outer end wall near the aperture may also be angled/beveled and terminates generally at the aperture. The shape of the channel is angled relative to a line of radius extending from the axis and towards an aperture, resulting in an angle that is ahead of the line of radius. In between the

apertures **434** are outwardly extending flaps or paddles **437**. The paddles **437** extend away from the disc **430** and separate the apertures. The paddles are spaced generally such that the brush wheel **452** is able to fit between subsequent paddles to remove a seed that is positioned between the paddles. However, the paddles **437** need not be included in all embodiments.

[0109] Also shown in the figures, and in more detail in FIG. **39**, is a singulator **500**. As noted, the singulator **500** is configured to singulate seed on the disc **430** so that a single seed is positioned at each aperture **434**. The singulator **500** is positioned to be attached to the housing **422** and “ride” along an outer edge **440** of the disc **430**. The singulator has a housing or body **504** that includes one or more surfaces for moving along the outer edge **440**. However, the singulator may or may not actually touch the disc **430**. The singulator **500** is attached to the housing **422** via member **502**, which may be spring steel to bias the singulator **500** away from the housing and towards the disc **430** in an axial direction. In addition, a biasing member **501** is included in the form of a spring wire to allow the singulator to float in a radial direction, which biases towards the axis of the disc **430**. The singulator further includes a blade **506** extending downward from the body **504**. A singulating element **508** extends from the blade **506**. In the figures, the singulating element is in the form of a plurality of offset and spaced brushes, which are configured to face the seed side **431** of the disc **430** about the seed path to “brush” off double seeds that may adhere to a single aperture. The brushes could be replaced with blade members, as disclosed in U.S. Pat. No. 9,277,688, which is hereby incorporated by reference in its entirety.

[0110] Additionally, as shown in FIG. **37** and disclosed herein, the disc **430** includes an inner portion **438** that is substantially planar, and an outer portion **439** that is curved. The curved outer portion **439** begins approximately the location of the seed apertures **434** and includes a curvature similar in nature to the outer profile of the brush wheel **452**. The curved outer portion **439** terminates at an edge **440**. It is also noted that the figure shows that the outer profile/edge **440** extends outward of the outer edge **441** of the second side **432** of the disc **430**.

[0111] FIG. **40** is a view of the second side **432** of the disc **430**. As shown, the apertures **434** extend through the disc **430**, and have a profile on the second side in which the aperture is larger on the second side and narrows towards the first side of the disc. This allows the pressure to be increased when felt at the first side **431** of the disc **430**. The gear teeth **435** are also shown in greater detail in FIG. **40**, as is the knock-off or ejector member **424**. As noted, the ejector **424** is a rolling type that includes a rotatable body **443** with extensions **444** radially spaced therefrom. The spacing of the extensions **444** coincide with the apertures **434**. The body **443** is connected to an arm **445** that is hingeably connected to the housing **422** at a hinge **425**. This allows for biasing of the ejector **424** towards the disc **430**, while allowing for axial movement of the same, which can be needed if obstructions are included, or if the disc **430** becomes warped or otherwise deformed.

[0112] The figure also shows the singulator **500** attached to ride along the edge **440** of the disc **430**.

[0113] In addition, the figure, along with FIGS. **41-42**, disclose the inclusion and use of a seal or gasket **480** on the second side **432** of the disc **430**. The seal or gasket **480** is used to create a fully pressurized zone on the second side of the disc **430**, making it so the area inside the seal or gasket **480**, which includes the entirety of the seed path, is pressurized. This ensures that all of the apertures **434** will be under pressure at all times of operation of the seed meter. The seal is housed in a notch or compartment **487** (see, e.g., FIG. **30**) that is formed between a portion of the housing **422** and an interior of the outer edge or wall **441** of the disc **430**.

[0114] The seal or gasket **480** comprises a resilient, pliable, or otherwise flexible material (such as a rubber, silicone, or like material) that will close off the portion of the disc **430** radially internal of the seal **480** to create a pressurized zone. To aid in creating such a zone, the seal **480** includes a unique design, as shown in FIG. **42**. The seal **480** includes a base portion **483** and a lip **485** extending therefrom. The lip **485** is curved and extends generally alongside the base **483**, and is connected at a hinge **486**. The composition of the seal or gasket **480** and the use of the hinge **486** allows for the base **483** and the lip **485** to move relative to one another without breaking or

fracturing. As the pressure is turned on and off, the seal will be acted upon and will deform to close off the edges **481**, **482** on opposite sides, creating the closed off pressurized zone. For example, as the pressure acts on the seal, the lip **485** will be forced outwardly radially and the base **483** will be forced inwardly radially. The hinge **486** allows for the base **483** and lip **485** to move towards and away from one another.

[0115] Additional changes and/or variations may be made to the systems as shown and described. For example, a seed meter system, as shown and described in U.S. application Ser. No. 15/343,342, which is hereby incorporated by reference in its entirety, could be used in place of and with any of the seed delivery systems shown and described herein. The brush wheel and belt could receive the seed from the seed meter of the '342 application and control the delivery at the speed desired to match the ground speed of the planter and/or row unit.

[0116] Furthermore, it should be noted that any of the components, embodiments, aspects, systems, or portions of any of the figures as shown and/or described could be used with any of the other the components, embodiments, aspects, systems, or portions of any of the figures as shown and/or described to result in additional embodiments. Those skilled in the art would readily understand and know, without undue testing, to replace the components based upon the information disclosed herein. This includes, but is not limited to, the type of pressure, the number of discs of a seed meter, the number of seed meters of a row unit, the use of sensors, the positioning of the meters and/or delivery systems relative to the direction of travel of the planter, the source of pressure (single source with hoses or integral), the source of motor, use of downforce, etc.

[0117] The embodiments and aspects of the invention as shown and described provide numerous advantages. The controlled nature of the delivery of seed from a seed meter to the furrow provides for increased accuracy and/or efficiency in spacing of seeds planted, especially with regard to higher speed planting. It is noted that high speed planting is recognized as planting at speeds generally above 8-miles per hour, and may be considered generally above 8-MPH and between 8-15 MPH (approximately 12.9-24.1 Kilometers per hour). However, this is not to be limiting, and high speed planting may be defined as being outside of this range. The high speed ability to plant allows farmers to plant their crop in reduced time and to plant within a preferred window of time based, at least in part, by the geographical climate and conditions of the location of planting. The aspects disclosed will allow for such high speed planting, while maintaining a high rate of singulation and planting efficiency and accuracy, in terms of ideal spacing and population of crop planting by the systems provided.

[0118] Thus, various configurations of seed delivery systems have been shown and described. It should be appreciated that the systems shown and described are for exemplary purposes, and the invention of a controlled system for delivering seed from a singulating seed meter to the ground to provide for consistent and equidistant spacing of the seed in the ground has thus been provided. It is to be contemplated that numerous variations, changes, and otherwise, which are obvious to those skilled in the art are to be considered part of the present invention.

Claims

1. An agricultural planting implement, comprising: a plurality of row units, each of the row units comprising: a seed meter, said seed meter including a housing and a seed disc therein, the housing having a seed side and a vacuum side, wherein the vacuum side applies a negative pressure to a full seed side of the seed disc; and a seed conveyor in communication with the seed meter to receive seeds from the seed meter, wherein the seed conveyor is adapted to move the seed towards an ejection location proximate to a bottom of a furrow and is adapted to eject the seed with substantially no horizontal velocity relative to the bottom of the furrow.
2. The agricultural planting implement of claim 1, further comprising a knock-off member on the vacuum side of the meter housing, the knock member including an extension that extends at least

partially into a seed aperture of the seed disc to dislodge a seed held in the seed aperture by the negative pressure.

3. The agricultural planting implement of claim 2, wherein the knock-off member comprises a rotating member with a plurality of extensions extending outwardly from the rotating member.

4. The agricultural planting implement of claim 2, wherein the knock-off member further comprises a hinge to connect the knock-off member to the seed meter housing.

5. The agricultural planting implement of claim 4, wherein the hinge is biased to urge the knock-off member towards the seed disc.

6. The agricultural planting implement of claim 1, wherein the seed disc comprises a plurality of seed apertures radially spaced to form a seed path and a curved portion radially external the seed path.

7. The agricultural planting implement of claim 1, further comprising a seal positioned inside an outer edge of the vacuum side of the seed disc and extending circumferentially about the outer edge of the disc to create a fully pressurized zone within the seal on the vacuum side of the seed disc.

8. The agricultural planting implement of claim 1, further comprising a seed carrier in communication with a seed exit of the seed meter, said seed carrier receiving seed from the seed meter one seed at a time.

9. The agricultural planting implement of claim 8, wherein the seed carrier receives seed that has been dislodged by the knock-off member.

10. The agricultural planting implement of claim 9, wherein the seed carrier is a brush wheel.

11. The agricultural planting implement of claim 1, wherein the seed conveyor comprises a flighted belt.

12. A method of planting seed, comprising: adhering seed to a seed aperture of a seed disc using a negative pressure, wherein said seed disc having a plurality of seed apertures forming a seed path; without cutting off the negative pressure, dislodging the seed from the seed aperture; receiving the dislodged seed with a carrier member and moving the seed via the carrier member in a rotational manner; transferring the seed from the carrier member to a seed conveyor towards the ground and releasing the seed from the seed carrier with substantially no horizontal velocity relative to the ground.

13. The method of claim 12, wherein the carrier member comprises a rotating brush.

14. The method of claim 13, wherein the seed disc rotates about a seed disc axis that is generally aligned with a direction of travel.

15. The method of claim 14, wherein the rotating brush rotates about a brush axis that is generally horizontal and transverse to the seed disc axis.

16. The method of claim 12, wherein the conveyor comprises a flighted belt within a conveyor cover.

17. An agricultural planting implement including a plurality of row units, each of the plurality of row units comprising: a seed meter comprising a seed meter housing; a rotatable seed disc within the seed meter housing and separating the seed meter housing into a seed side and a vacuum side, said seed disc comprising a plurality of seed apertures radially spaced to form a seed path; and a seed conveyor in communication with the seed meter to receive seeds from the seed meter, wherein the seed conveyor is adapted to move the seed towards an ejection location proximate to a bottom of a furrow and is adapted to eject the seed with substantially no horizontal velocity relative to the bottom of the furrow.

18. The agricultural planting implement of claim 17, further comprising a knock-off member on the vacuum side of the meter housing, the knock member including an extension that extends at least partially into a seed aperture of the seed disc to dislodge a seed held in the seed aperture by a negative pressure, wherein the knock-off member comprises a rotating member with a plurality of extensions extending outwardly from the rotating member.

19. The agricultural planting implement of claim 18, wherein the plurality of extensions of the

knock-off member are configured to align with the seed apertures of the seed disc as the knock-off member rotates.

20. The agricultural planting implement of claim 17, further comprising a seal positioned inside an outer edge of the vacuum side of the seed disc and extending circumferentially about the outer edge of the disc to create a fully pressurized zone within the seal on the vacuum side of the seed disc.
